

Development and characterization of a phenotype algorithm to identify Acute Kidney Injury in RWD

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Background

Acute kidney injury (AKI) affects 1 in 5 adults and 1 in 3 children worldwide during a hospital episode of care according to a comprehensive meta-analysis using the KDIGO definition referred below (1). Epidemiological evidence suggests that even mild, reversible AKI has important clinical consequences including an increased mortality. Therefore, many efforts have been made over the last two decades to reach a consensus on the AKI definition and to raise awareness to improve diagnosis, prevention, and treatment. We aimed to develop and characterize a phenotype (cohort definition) to identify patients with AKI in electronic health records or claim databases.

Methods

In the context of the OHDSI “*Phenotype Phebruary*” 2022 initiative (<https://www.ohdsi.org/phenotype-phebruary-2022/>), we implemented best practices to generate a valid phenotype for AKI, as follows: First, we conducted a literature review to elaborate a relevant clinical description of the entity, AKI. Second, we explored previous examples of cohort definitions published in the literature with a focus on codes lists used, algorithms implemented and, were available, validation analyses. Finally, we used ATLAS to generate concepts sets and cohort definitions based on the extracted literature and clinical description. This resulted in two concepts sets (sensitive and specific), and four cohort definitions, using the sensitive and specific concept sets combined with and without laboratory values of serum creatinine. These four cohorts were subsequently evaluated using the web-based ATLAS tool and CohortDiagnostics from HADES, a collection of open-source R packages, applied to OMOP CDM-converted instances of Hospital del Mar-IMASIS, MarketScan, Optum claims, and Optum EHR.

Results

Literature review and phenotype building

Acute Kidney Injury is one of many acute kidney diseases and disorders (AKD). According to international guidelines (2), AKI is a common, harmful and potentially treatable disease defined by an abrupt decrease in kidney function including, but not limited to acute kidney failure. The etiology of AKI encompasses specific kidney diseases (acute interstitial nephritis, acute glomerular and vasculitic renal diseases); non-specific conditions (ischemia, toxic injury); and extrarenal conditions (prerenal azotemia, acute postrenal obstructive nephropathy). The clinical presentation of AKI is similar regardless of its etiology; thus, we focused on a broad phenotype to ascertain the full range of AKI subtypes. Differential diagnosis of AKI is shown in Table 1 (2,3).

Table 1. Differential diagnosis of acute kidney injury (AKI), acute kidney disease (AKD), chronic kidney disease (CKD) and no kidney disease (NKD) (adapted from international clinical guidelines, Kellum et al. 2012).

	AKI	AKD	CKD	NKD*
Duration	≤7 days	<3 months	>3 months	NA
Functional criteria	Increase in *sCr by ≥50% within 7 days or increase in sCr by ≥0.3 mg/dl (26.5 μmol/l) within 2 days or oliguria for ≥6 hours	AKI or **GFR <60 ml/min/1.73 m ² or decrease in GFR by ≥35% over baseline or increase in sCr by >50% over baseline	GFR < 60 ml/min/1.73 m ²	GFR ≥ 60 ml/min/1.73 m ² , stable GFR (no decrease by 35% within 3 months), stable sCr (no increase by 50% within 3 months or increase by 0.3 mg/dl within 2 days), no oliguria for ≥6 hours
AND/OR	OR	OR	OR	AND
Structural criteria	Not defined	Elevated marker of kidney damage (albuminuria, haematuria or pyuria are most common)	Elevated marker of kidney damage (albuminuria is most common)	No marker of kidney damage

*sCr: serum creatinine, **GFR: glomerular filtration rate

Multiple observational studies using databases proposed lists of diagnostic codes to identify patients with AKI (ICD-10 and ICD-9, mostly)(4–7). Based on those codes, we generated a broad concept set with the following concepts related to AKI (sensitive concept set): acute renal failure syndrome, acute nephritis, and rapidly progressive nephritic syndrome, and a narrow concept set of AKI (specific concept set) including one concept: acute renal failure syndrome. In both instances, all descendant concepts were explored and added to each set. One cohort definition was developed using the sensitive concept set, and one using the specific concept set. Cohort exit was defined 7 days after index date, as loss of renal function persisting more than 7 days is considered a condition different from AKI (Table 1). In addition, the following inclusion criteria were added to both cohort definitions: 1) A minimum of 1-year prior observation, 2) No end-stage kidney disease, in the year prior, 3) No chronic dialysis, 4) A washout period of no AKI in the 30 days preceding the entry date (to prevent follow-up clinical encounters related to the same AKI event as opposed to a new AKI event).

Variation of measurements relative to a baseline (deltas and ratios) are not supported as an entry event in ATLAS, thus clinical criteria based on serum creatinine (sCr) variation with respect to a baseline, cannot be established in ATLAS cohort definitions without flawed workarounds. Besides,

availability of data on sCr with harmonized units within the database, was only found in one dataset (CPRD, results not shown). The implementation of the laboratory measurements workaround in this dataset, rendered very few additional patients to the already identified based on the concept sets alone. Consequently, we decided to use condition occurrences as the only domain of entry events, as no solution seemed satisfactory to add laboratory values, resulting in two cohort definitions to be characterized and evaluated.

Cohort Diagnostics

We characterized the two selected phenotypes in all four databases to evaluate each definition and generate insights on potential divergencies. Table 2 shows the main patient characteristics of each cohort per dataset. We observed a substantial overlap using either phenotype in both databases, with 0.3% incremental sample size for Optum EHR, 0.4% for Optum claims and 1% for MarketScan. Incidence rate by age, calendar year and sex showed a consistent order of magnitude and trend as expected from the literature. Within database differences in medical conditions at baseline between both phenotypes were not higher than 0.3% in any case. Likewise, observation time was almost identical within databases, ranging from 2585 of average prior observation days in Optum EHR, to 1812 days in Optum Claims.

A complete characterization and evaluation of the phenotype will be presented at the symposium.

Table 2. Demographic characteristics of AKI patients identified using two distinct phenotypes (broad and narrow) in four databases from Spain and the US

	Marketscan				Optum Claims				Optum EHR				Hospital del Mar			
	Broad		Narrow		Broad		Narrow		Broad		Narrow		Broad		Narrow	
N	1,457,733		1,441,618		2,793,226		2,781,688		3,274,720		3,259,455		21,330		21,292	
Age (years), mean SD	66.42	17.4	66.66	17.18	73.57	12.56	73.64	12.45	66.63	15.6	66.73	15.48	78.41	12.93	78.45	12.84
Age, 5-year categories, n %																
20 - 24	18,319	1.26%	17,637	1.22%												
25 - 29	14,687	1.01%	14,132	0.98%					40,491	1.24%	39,827	1.22%				
30 - 34	19,458	1.33%	18,723	1.30%					53,838	1.64%	53,178	1.63%				
35 - 39	28,633	1.96%	27,712	1.92%					68,267	2.08%	67,545	2.07%				
40 - 44	43,855	3.01%	42,715	2.96%	28,887	1.03%	28,355	1.02%	90,025	2.75%	89,238	2.74%				
45 - 49	67,779	4.65%	66,502	4.61%	47,355	1.70%	46,698	1.68%	132,315	4.04%	131,377	4.03%	328	1.54%	326	1.53%
50 - 54	104,761	7.19%	103,282	7.16%	77,961	2.79%	77,279	2.78%	198,842	6.07%	197,734	6.07%	466	2.18%	463	2.17%
55 - 59	156,096	10.71%	154,361	10.71%	130,320	4.67%	129,462	4.65%	282,961	8.64%	281,759	8.64%	689	3.23%	685	3.22%
60 - 64	220,416	15.12%	218,622	15.17%	190,539	6.82%	189,559	6.81%	345,120	10.54%	343,875	10.55%	946	4.44%	945	4.44%
65 - 69	104,048	7.14%	103,381	7.17%	324,032	11.60%	322,950	11.61%	365,207	11.15%	363,985	11.17%	1,339	6.28%	1,332	6.26%
70 - 74	127,898	8.77%	127,205	8.82%	456,450	16.34%	454,976	16.36%	382,163	11.67%	381,115	11.69%	2,025	9.49%	2,025	9.51%
75 - 79	141,923	9.74%	141,383	9.81%	460,537	16.49%	459,248	16.51%	428,657	13.09%	427,693	13.12%	2,913	13.66%	2,911	13.67%
80 - 84	150,708	10.34%	150,284	10.42%	456,871	16.36%	455,971	16.39%	621,165	18.97%	620,372	19.03%	4,062	19.04%	4,059	19.06%
85 - 89	135,589	9.30%	135,395	9.39%	464,932	16.65%	464,464	16.70%	208,356	6.36%	208,236	6.39%	4,481	21.01%	4,479	21.04%
90 - 94	79,198	5.43%	79,066	5.48%	99,767	3.57%	99,728	3.59%					2,770	12.99%	2,768	13.0%
95 - 99	22,114	1.52%	22,095	1.53%									734	3.44%	734	3.45%
Race, n %																
Asian	NA	NA	NA		63,279	2.27%	62,696	2.25%	36,412	1.11%	35,811	1.10%	NA		NA	
Black or African American	NA	NA	NA		435,282	15.58%	434,156	15.61%	562,799	17.19%	561,494	17.23%	NA		NA	
White	NA	NA	NA		1,872,383	67.03%	1,864,857	67.04%	2,495,231	76.20%	2,483,247	76.19%	NA		NA	
Sex, n %																
Female	642,580	44.08%	634,235	43.99%	1,357,025	48.58%	1,350,789	48.56%	1,547,458	47.25%	1,539,465	47.23%	10,094	47.32%	10,076	47.32%
Male	815,153	55.92%	807,383	56.01%	1,436,201	51.42%	1,430,899	51.44%	1,727,262	52.75%	1,719,990	52.77%	11,236	52.68%	11,216	52.68%
Pre-index observation time (days), mean SD	2195.52	1519.58	2199.69	1521.5	1812.4	1204.25	1813.04	1204.80	2585.29	1216.82	2586.62	1217.2	6424.05	3066.42	6426.95	3065.53
Post-index observation time (days), mean SD	569.17	595.71	565.59	592.08	679.21	733.00	677.49	731.49	882.60	795.21	880.1	793.76	658.89	798.83	657.55	797.36

Conclusion

Our proposed phenotype (cohort definition) for AKI is useful to build epidemiological cohorts of patients presenting this severe condition in EHR or claims databases with and without access to lab measurement values. Using the current version of ATLAS for cohort definitions that entail delta or ratio of laboratory measurements as an entry event is not feasible. This may affect sensitivity of our phenotype across datasets, but likely not specificity. Between the narrow and broad cohort definitions there is little incremental value in the databases explored, thus we recommend using the narrow version. A complete evaluation of the phenotype will be presented at the symposium.

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